THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The charge on an electron is: 1

(a) variable **(b)** negative **(c)** zero **(d)** positive

2. One breath is about 0.5 litre and a person takes about 20000 breaths a day. The air breathed in a day is about: 1

(a) 100 litres **(b)** 10000 litres **(c)** 1000 litres **(d)** 100000 litres

3. Which of the following is a colloid? 1

(a) sugar in water **(b)** air **(c)** fog **(d)** salt in water

4. The atomicity of sulphur (S_8) is: 1

(a) 1 **(b)** 2 **(c)** 8 **(d)** 4

5. The mass number of an atom equals the number of: 1

(a) protons plus neutrons **(b)** protons only
(c) neutrons plus electrons **(d)** protons plus electrons

6. The conservation of energy is an example of: 1

(a) a symbol
(b) an estimate
(c) a general principle that applies in every process
(d) a unit

7. In chromatography the component which is more soluble in the solvent: 1

(a) moves farther up the paper **(b)** moves down
(c) disappears **(d)** stays at the base

8. The chemical formula of sodium chloride is: 1

(a) $NaCl$ **(b)** Na_2Cl_2 **(c)** Na_2Cl **(d)** $NaCl_2$

9. An atom with a complete outermost shell is generally: 1

(a) positive **(b)** radioactive
(c) very reactive **(d)** chemically unreactive

10. A scientific law usually: 1

(a) explains why the pattern exists in every detail
(b) is a personal opinion
(c) cannot be tested
(d) describes a regular pattern observed in nature

11. Which of the following is a homogeneous mixture? 1

(a) brass **(b)** sand and sugar **(c)** oil and water **(d)** muddy water

12. According to the law of conservation of mass, in a chemical reaction the total mass of the reactants: 1

(a) is zero **(b)** equals the total mass of the products
(c) is less than the products **(d)** is greater than the products

13. The nucleus of an atom was discovered by: 1

(a) Thomson **(b)** Rutherford **(c)** Bohr **(d)** Dalton

14. The gap between a prediction and observation is an opportunity to: 1

(a) refine the model **(b)** blame the instrument
(c) change the data **(d)** stop experimenting

15. Colloidal particles pass through ordinary filter paper because they are: 1

(a) heavier than water **(b)** larger than the pores of the paper
(c) magnetic **(d)** smaller than the pores of the paper

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is centrifugation? Give one use. 2

17. What is atomicity? Give the atomicity of H_2, O_3 and P_4. 2

18. Calculate the number of protons, neutrons and electrons in $^{23}_{11}Na$. 2

19. What do scientists do when a prediction does not match an observation? 2

20. Calculate the mass by mass percentage of a solution having 25 g of sugar in 75 g of water. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Differentiate between isotopes and isobars (three points). 3

22. Calculate the mass of 2.5 litres of water in kg and in g (density 1 kg per litre). 3

23. What is meant by a saturated solution and an unsaturated solution? How can we convert one to the other? 3

24. Differentiate between an atom and a molecule (three points). 3

OR

Explain the three isotopes of hydrogen with their composition. 3

25. Describe how an estimate of the number of litres of air breathed in a day can be made. 3

OR

Describe how crystallisation is used to purify an impure sample of copper sulphate. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain, with examples, why measurements are never perfectly exact, and how errors can be reduced. 5

OR

Describe the working and use of paper chromatography with the diagram described in words, including R_f ideas. 5

27. A compound contains carbon and oxygen in the ratio 3 : 8. Find the mass of the compound formed when 18 g carbon is burnt and the mass of oxygen needed. Name the compound. 5

OR

Explain how the atomic number and the mass number identify an atom. Complete the table for ${}^{39}_{19}\mathrm{K}$, ${}^{32}_{16}\mathrm{S}$ and ${}^{19}_{9}\mathrm{F}$. 5

END OF QUESTION PAPER — **Total: 50 Marks**

MASTERMIND